

**Comgin back to
scRNA-seq data...**

So far we...

- Removed outlier cells
 - Too low expression (damaged)
 - Too high expression (doublets)
 - Too much mitochondrial genes (stress or damage)
- Removed outlier genes
 - Expressed in less than 3 cells

The result is still a very sparse matrix, but without the most egregious technical artifacts.

Normalization

- "Standard library size" of 10.000 transcripts per cell.
- Why?
 - Measurement error: Some cells are simply measured more precisely than others.
- Raw count $\Rightarrow$ Relative abundance w.r.t. an "ideal" cell that would have 10.000 transcripts.
- If we did not do normalization, many statistical tools used to compare individual cells would be biased to pick up the good/bad cell signal instead of actual differences in expression.
- Downside: Some cell types are naturally "small" or "large". This transformation erases that---for some analysis, this may be what we want, but not always!

Log1p transformation

- Map every normalized count x to $\log_2(x + 1)$.
- Why? Variance stabilization.
 - Gene expression variance correlates with mean expression $\Rightarrow$ Highly expressed genes are much more variable.
 - If we wanted to look at genes with high variability, standard tools would simply report all high expression genes.
 - High expression is not always a good indicator of usefulness (e.g., housekeeping genes).
 - Applying logarithm "compacts" the range, making low-expression high-variability genes stand out more compared to high-expression low-variability.

Log1p transformation

- Map every normalized count x to $\log_2(x + 1)$.
- Why? Multiplicative vs. additive comparisons.
 - Kind of similar principle as before, but worth bringing up.
 - Changes in gene expression are often multiplicative: Going from 10 to 20 transcripts is almost as equally significant as going from 1000 to 2000.
 - After applying the logarithm, "multiplicative differences" are now "additive differences".
 - $\log_2(10) + 1 = \log_2(20)$
 - $\log_2(1000) + 1 = \log_2(2000)$
 - We could correct for this within our statistical methods, but this transformation ensures it happens consistently.
 - Still, we have to check which methods expect normalized and log-transformed data and which do not!

Log1p transformation

- Map every normalized count x to $\log_2(x + 1)$.
- Why? Reducing distribution skew.
 - Raw gene expression is typically not normally distributed---it is skewed towards higher values (if we don't count zeros).
 - Log-transform corrects that skewness $\Rightarrow$ We can now (mostly) assume that the data for each gene is normally distributed.
- Why not?
 - Amplifies noise and variance in low expression genes.

Highly variable genes selection

- Only retain genes that (after normalization / log-transform) have the highest variance.
- Simple rationale: If we want to find differences between multiple cell types, genes that are constant along all cells are probably not contributing to that difference.
- Extreme case: Filtering out genes that are not expressed at all is clearly "safe" for our goal.
- Disadvantages: There is generally no good way to tell how many genes we should keep. Depending on how strict we are, we may miss some genes that are crucial for small sub-populations of cells (e.g., there are 100/100.000 that we actually want to detect).
- Bottom line: HVGs retain the "most visible" differences between cell types, but not always the most consequential ones.

Quickly about PCA, UMAP,
Batch correction...

Data discretization hands on

Moving back to bonesis...